

1. Image Enhancement is the process of improving the visual appearance of an image by directly modifying the pixel values. In the spatial domain, operations are performed on individual pixels or their neighboring pixels.

Techniques used :-

1. Gray-level transformation :- It changes the intensity values of pixels to improve image visibility.

Types $\Rightarrow$

- Image Negative
- Log transformation
- Power-Law (gamma correction)
- Contrast stretching

2. Histogram processing :- It represents the distribution of pixel intensities.

Methods $\Rightarrow$

- Histogram Equalization
- Histogram Equilization

3. Spatial filtering :- It uses a small mask (kernel) to modify pixel values.

Types $\Rightarrow$

- Smoothing filters (Mean, Median)
- Sharpening filters (Laplacian, Sobel)

2.

Image Smoothing :- It reduces noise and small variations in an image.

working principle :- A smoothing filter replaces each pixel with the average or median value of its neighboring pixels.

Advantages :-

- Removes random noise
- produces a clearer image.
- Improve image quality.

Applications :-

- Medical images
- Satellite images
- CCTV images

Image sharpening :- It increases edge details and fine structures.

working principle :- Sharpening filters highlight intensity changes between neighboring pixels.

Advantages :-

- Enhance boundaries
- Improves feature visibility.

Applications :-

- character recognition
- Face detection

Difference b/w Smoothing and sharpening :-

Image Smoothing

1. Removes noise
2. Blur image slightly
3. Uses averaging.
4. Applied before Segmentation.

Image sharpening

1. Enhance edges
2. Increases sharpness
3. Uses derivatives.
4. Applied before feature extraction.

3. Frequency domain enhancement processes an image after converting it into the frequency domain using the fourier transform.

Steps:

1. Convert image using Fourier transform.
2. Apply frequency filter.
3. perform inverse fourier transform.
4. Obtain enhanced image.

Low-pass filter :- It allows low-frequency components to pass and removes high-frequency components.

Effects

- Smoothens image
- Reduces noise
- Removes unwanted details

Applications

- Medical imaging
- Noise reduction
- Image preprocessing

High-pass filter :- It allows high-frequency components and suppresses low frequencies

Effects

- Enhances edges
- Improves sharpness

Applications

- Object detection
- Fingerprint recognition.

Comparison :-

Low-pass filter

1. Removes noise
2. Smoothens image
3. passes low frequencies

High-pass filter

1. Enhance edges
2. sharpens image
3. passes high frequencies.

4. Image segmentation divides an image into meaningful regions so that objects can be analyzed separately.

1. point detection :- It detects isolated pixels that differ significantly from neighboring pixels.

Advantages

- Simple
- Detects isolated defects

Limitations

- sensitive to noise.

2. Line detection :- It detects horizontal, vertical, diagonal lines using masks.

Advantages

- Detects roads & boundaries

Limitations

- Less effective on curved objects.

3. Edge detection :- It detects sudden intensity changes.

Advantages

- Accurate boundary detection

Limitations

- sensitive to noise.

4. Thresholding :- It separates foreground and background using intensity values.

Advantages

- Fast
- Easy implementation

Limitations

- poor performance under uneven lighting.

5. Region-based segmentation :- It groups neighboring pixels with similar properties.

Advantages

- produces complete regions

Limitations

- computationally expensive

5. Edge detection :- Detects sudden changes in pixel intensity using gradient operators.

Importance

- Identifies object boundaries
- Simplifies image analysis

Application

- Face recognition
- Medical imaging

Edge Linking :- Connects broken edge pixels into continuous edges using similarity in direction and distance.

Importance

- produces complete object outlines.
- Reduces fragmented edges

Applications

- shape analysis
- OCR
- Industrial inspection

Boundary detection :- It extracts the outer boundary of segmented objects.

Importance

- Measures objects size
- Supports defect detection

Applications

- cell analysis
- Machine vision.

Contribution to object Recognition :-

- Detects important object features.
- Improves classification accuracy.
- Helps distinguish different objects.

